

Multimodal Data Integration Improves Disease Risk Prediction in the UK Biobank

Xiayuan Huang, PhD¹, Hang Zhou, PhD¹, Yitao Hong, BS¹, Xin Zhou, PhD¹, Johann de Jong, PhD^{2,*}, Zuoheng Wang, PhD^{1,*}

¹Yale University, New Haven, CT, USA

²UCB Biosciences GmbH, Monheim am Rhein, Germany

*Equal contribution as co-senior authors

Abstract

Family health history is an important component to assess risk for common chronic diseases. The integration of electronic health records and genetic data offers great potential to improve disease risk prediction by capturing both clinical and genetic risk factors. We present ALIGATEHR-Gen, a graph attention network that integrates multimodal patient data including genetic information, diagnosis codes, and demographics, along with external medical ontology knowledge. ALIGATEHR-Gen constructs unified patient representations by incorporating genetically inferred first-degree relationships and disease ontology embeddings to enhance disease risk prediction. We evaluate the predictive performance of ALIGATEHR-Gen across 118 diseases in the UK Biobank and demonstrate that it outperforms state-of-the-art baseline models by an average of at least 6%. A case study on five primary fibrotic and closely related diseases reveals that ALIGATEHR-Gen effectively distinguishes patient subgroups based on clinical and genetic features. These findings illustrate the potential of ALIGATEHR-Gen to advance predictive and interpretable modeling in healthcare.

Introduction

The modern healthcare system has been transformed by the proliferation of diverse data sources and multimodal patient information^{1,2}. Advances in digital health technologies and biomedical research have fostered a data-rich environment where electronic health records (EHR) and genetic data offer unprecedented opportunities to advance genomic medicine. EHRs serve as digitized repositories of patients' medical histories, including consultations, diagnoses, procedures, laboratory tests, and prescribed medications, while genetic data provides a comprehensive view of patients' genomic profiles, enabling the assessment of genetic contributions to disease risk³.

The integration of multimodal data in EHR has facilitated the development of machine learning-based disease risk prediction models⁴. In particular, deep learning architectures, such as recurrent neural networks (RNN)^{5,6}, graph neural networks (GNN)⁷, and transformers⁸, have demonstrated great potential in enhancing diagnostic predictions from EHR, suggesting the power of multimodal data integration in predictive healthcare analytics. Despite the considerable progress in predictive modeling across data modalities in EHR⁹, the use of genetic information into clinical risk assessment remains limited. The convergence of EHR and genetic data presents a unique opportunity for a synergistic approach to patient modeling, where clinical insights from EHR are enriched by genetic contributions to disease susceptibility. Integrating these data sources into patient modeling holds great promise for enhancing disease risk prediction, improving patient stratification, and deepening our understanding of genetic influence on health outcomes.

Family health history is an important component to assess risk for common chronic diseases, because families share genetic variations, environmental exposures, common lifestyle, and health-related social factors. Unfortunately, family health history is often missing or inconsistently recorded in EHR databases¹⁰. Thus, patient modeling frequently overlooks this valuable information, potentially reducing both predictive accuracy and clinical utility. To address this limitation, we previously developed ALIGATEHR, which models the inferred family relations from de-identified demographic data in a graph attention network augmented with an attention-based medical ontology representation¹¹. By accounting for the complex influence of genetics, shared environmental exposures, and disease dependencies, ALIGATEHR enhances patient representation learning. Its application to real-world healthcare data demonstrates that integrating inferred family relations improves disease risk prediction.

Large-scale biobanks, such as the UK Biobank (UKB)¹², have become invaluable resources for investigating the role of genetics and its interplay with other factors in human health. By linking decades of EHR with genetic data, these biobanks provide a robust platform for assessing the risk of common diseases. The availability of genome-wide single nucleotide polymorphism (SNP) data enables the identification of familial relationships among participants, which is important for epidemiological analyses¹³ and genome-wide association studies (GWAS)¹⁴⁻¹⁶. For example,

approximately 30% of UKB participants have been inferred to have third-degree relatives or closer in the cohort. These genetically inferred relationships provide unprecedented opportunities to study the risk of heritable diseases in real-world settings. By leveraging this information, we can systematically reconstruct family health history to enhance disease risk prediction. Additionally, incorporating genetic data allows for more precise genetic analyses to improve clinical risk estimation and enabling personalized treatment strategies.

In this work, we introduce ALIGATEHR-Gen (ALigning Graph Attention neTworks for EHR using Genetic kinship), a generic framework for learning patient representations to enhance disease risk prediction by integrating EHR and genetic data. ALIGATEHR-Gen learns patient representations in a graph attention network to account for family health history based on genetically inferred relationships. To further enhance the quality of the learned representations by accounting for misdiagnosis and disease dependencies, we additionally integrate a medical ontology of embedded diagnosis codes into the attention mechanism. Applied to the prediction of disease risks, ALIGATEHR-Gen explicitly models genetically inferred relationships and improves the predictive performance across a broad spectrum of diseases.

The core contributions of our work are summarized as follows:

- We introduce an attention-based GNN framework, ALIGATEHR-Gen, for disease risk prediction, an important area in health informatics. To the best of our knowledge, it is the first attention-based GNN model that integrates clinical features and genetic data for disease risk prediction, using genetically inferred first-degree relationships as graph edges. By linking a patient's disease risk to the clinical profiles of their first-degree relatives, ALIGATEHR-Gen effectively captures genetic aspects of diseases. Additionally, its attention-based mechanism prioritizes the most influential first-degree relatives and identifies clinically relevant features for disease risk prediction.
- We demonstrate that ALIGATEHR-Gen outperforms state-of-the-art baseline models across a broad spectrum of diseases in a large-scale real-world biobank dataset, the UKB. Additionally, the ablation experiment confirms the robustness of our model.
- We show ALIGATEHR-Gen's ability to distinguish patient subgroups across five primary fibrotic or closely related diseases: coronary artery disease, chronic kidney disease, metabolic dysfunction-associated steatohepatitis, pulmonary fibrosis, and Crohn's disease. This real-world clinical use case highlights the importance of incorporating genetic information in predictive modeling and demonstrates ALIGATEHR-Gen's utility in enhancing patient representation learning for predictive and interpretable EHR analyses.

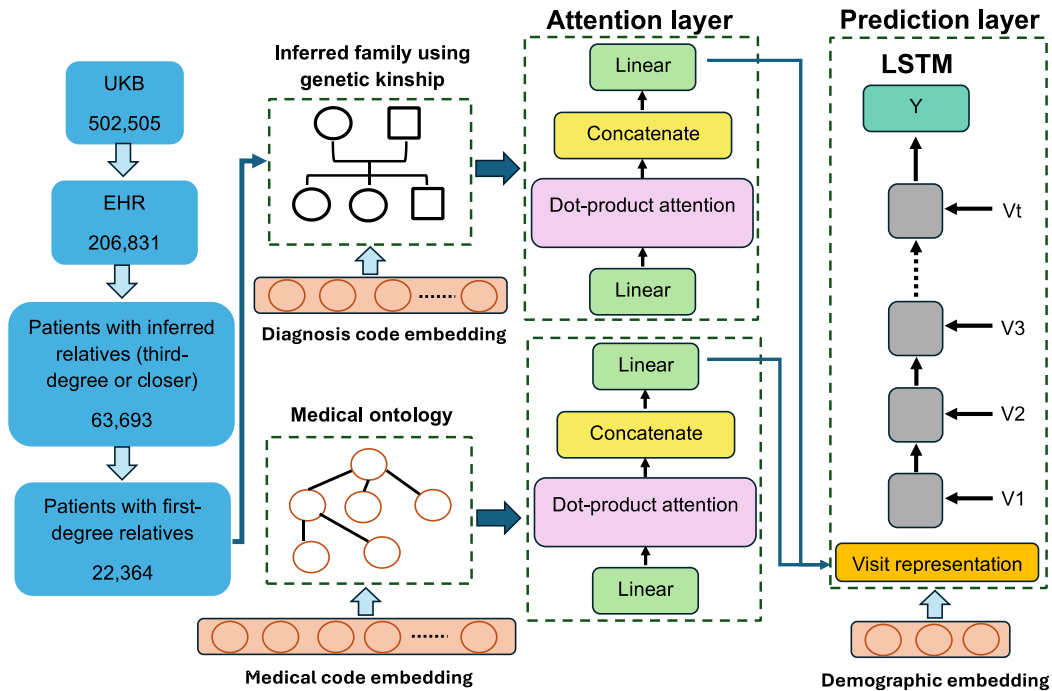


Figure 1. Illustrative diagram of ALIGATEHR-Gen.

Methods

Figure 1 presents an overview of the ALIGATEHR-Gen framework, which consists of four main components: data processing, graph construction, an attention layer, and a prediction layer. The data processing component extracts relevant clinical features from EHR and leverages genetic data to identify first-degree relatives. The graph construction step then builds two distinct graphs: a patient graph, in which edges link first-degree relatives, and an ontology graph, which encodes hierarchical relationships among medical concepts. In the attention layer, ALIGATEHR-Gen applies a graph-attention network¹⁷ to processing both the patient and ontology graphs. Finally, the visit representation, which integrates patient representation from genetically inferred relationships, ontology representation from the ontology graph, and patient demographic embeddings, serves as input for the prediction task.

Data Processing

We use data from the UKB, which includes over 500,000 participants. Among them, 206,831 have both de-identified EHR, including both primary care and hospital inpatient data, and genetic data. Primary care data, released in 2019, capture participants' diagnosis history up to 2016 or 2017, while hospital inpatient data provide information on hospital admissions through 2022. Genetic data were released in 2017. Within this cohort, 63,693 participants have genetically inferred third-degree or closer relatives, and 22,364 have genetically inferred first-degree relatives (Figure 1 and Table 1). A detailed description of the inference of close relatives using kinship coefficients is provided in the Patient Graph section. Based on this processed population, subsequent datasets are constructed for various predictive modeling tasks.

Table 1. Demographics of experimental population.

Statistics	UKB	CAD	CKD	MASH	PF	CD
Number of participants	22,364	5003	1275	220	456	71
Female	60%	33%	51%	46%	48%	53%
Male	40%	67%	49%	54%	52%	47%
Age at recruitment (mean)	56.9	61.3	62.1	58.3	56.8	56.3
Age at first diagnosis (mean) ¹	NA ²	64.6	69.9	64.6	69.5	60.2
Number of visits per patient (mean)	23.3	36.1	52.6	44.4	40.4	40.0
Number of unique ICD codes per patient (mean)	24.5	43.8	54.5	52.9	46.1	41.3
Number of ICD codes per visit (mean)	2.4	3.2	3.4	3.5	3.2	2.7

UKB: participants with first-degree relatives in UK Biobank; CAD: coronary artery disease; CKD: chronic kidney disease; MASH: metabolic dysfunction-associated steatohepatitis; PF: pulmonary fibrosis; CD: Crohn's disease.

¹Age at first diagnosis: a patient's age when initially diagnosed with the associated disease.

²Age at first diagnosis is not obtained for the general UKB population.

Graph Construction

ALIGATEHR-Gen constructs two graphs: a patient graph constructed from genetic kinship and an ontology graph derived from medical ontology.

Patient Graph

To characterize familial relatedness within the UKB participants, kinship coefficients are estimated by UKB for all pairs of samples using genome-wide SNP data^{12,18,19}. The kinship coefficient represents the probability that a randomly selected allele from one individual is identical by descent (IBD) with an allele at the same locus from another individual and provides a measure of relatedness between pairs of individuals. Kinship coefficients are recorded for pairs who are inferred to be third-degree relatives or closer (UKB field ID: 22021). First-degree relatives are defined as pairs with a kinship coefficient larger than or equal to 0.177¹⁹⁻²¹. Based on these inferred relationships, a patient graph is then built by connecting first-degree relatives within the cohort.

Ontology Graph

In medical ontologies, the hierarchy of various medical concepts is often represented through a parent-child relationship, with diagnosis codes serving as leaf nodes. The ontology graph is modeled as a directed acyclic graph

(DAG), in which the nodes constitute a set of medical concepts. In this structure, parent nodes represent more general medical concepts, while child nodes represent more specific subcategories. This hierarchical organization offers a multi-level perspective on medical concepts, allowing for a comprehensive understanding of the medical domain by accommodating concepts at varying levels of specificity and granularity.

Attention Layer

Patient Graph

In the patient graph, each node (patient) in the graph has an associated feature vector representing a patient's disease status, denoted as $h_i \in \mathbb{R}^C$, with $|C|$ being the total number of unique diagnosis codes in the EHR data. Between the nodes, an attention mechanism is used to link a patient's representation to the clinical profiles of their relatives. We formulate the patient's final disease representation as follows:

$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha'_{ij} W_e h_j \right),$$

where $h'_i \in \mathbb{R}^C$ denotes the final representation of patient i , σ applies a ReLU activation function, $\mathcal{N}_i$ contains the indices of patient i 's first-degree relatives, W_e is the weight matrix which is applied to each node for a shared linear transformation, and the attention coefficient α'_{ij} of patient graph is calculated by the following softmax function,

$$\alpha'_{ij} = \text{softmax}(e'_{ij}) = \frac{\exp(e'_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(e'_{ik})},$$

where $e'_{ij} = \alpha(W_e h_i, W_e h_j)$. In our experiments, the attention mechanism is a single-layer feedforward neural network. Here e'_{ij} indicates the importance of first-degree relative j 's disease features to patient i .

Ontology Graph

In the ontology DAG, each node is assigned a basic embedding vector $e_i \in \mathbb{R}^m$, where m represents the dimensionality. Then $e_1, e_2, \dots, e_{|C|}$ are the basic embeddings of the diagnosis codes $d_1, d_2, \dots, d_{|C|}$. Let $g_i \in \mathbb{R}^m$ denote the final representation of the diagnosis code d_i , as a convex combination of the basic embeddings of itself and its ancestors:

$$g_i = \sum_{j \in \mathcal{A}(i)} \alpha_{ij} e_j, \quad \text{where } \sum_{j \in \mathcal{A}(i)} \alpha_{ij} = 1, \quad \alpha_{ij} \geq 0, \quad \text{for } j \in \mathcal{A}(i).$$

Here $\mathcal{A}(i)$ are the indices of the code d_i and d_i 's ancestors, e_j is the basic embedding of the code d_j , and $\alpha_{ij} \in \mathbb{R}$, the attention weight on the embedding e_j when calculating g_i . The attention weight α_{ij} is calculated by the softmax function,

$$\alpha_{ij} = \frac{\exp(f(e_i, e_j))}{\sum_{k \in \mathcal{A}(i)} \exp(f(e_i, e_k))},$$

where $f(e_i, e_j)$ is a scalar value representing the compatibility between the basic embeddings of e_i and e_j , computed via the feed-forward network,

$$f(e_i, e_j) = u_a^T \tanh \left(W_a \begin{bmatrix} e_i \\ e_j \end{bmatrix} + b_a \right),$$

where $W_a \in \mathbb{R}^{l \times 2m}$ is the weight matrix for the concatenation of e_i and e_j , $b_a \in \mathbb{R}^l$ is the bias vector, and $u_a \in \mathbb{R}^l$ is the weight vector for generating the scalar value, with l denoting the dimension size of the hidden layer $f(e_i, e_j)$.

Prediction Layer

Following the attention layer, we construct the final representation v_t for each patient at visit t by aggregating the representation of the patient's diagnosis status infused with information from first-degree relatives, and demographic embedding including age and sex. The visit representation v_t is then used as input to a predictive model for predicting the target label Y_t using recurrent neural networks. In this study, we use long short-term memory (LSTM) networks²² for our experiments. The prediction task is to predict disease risk at the next visit, denoted as Y_{t+1} at timestep $t + 1$, given all the previous visit history up to the current timestep $v_1, v_2, \dots, v_t$.

Experimental Setting

In this study, we define the prediction of disease onset at a patient's next clinical visit as a binary classification task. To construct the dataset, individuals are selected from EHR records based on their first documented diagnosis of the target disease, identifying them as potential case patients for each disease model. A total of 118 predictive models are

developed, each corresponding to a unique disease code, to systematically assess the performance and robustness of ALIGATEHR-Gen across diverse disease prediction tasks. These 118 diseases are common diseases from ten ICD coding categories²³: “Diseases of the Blood and Blood-Forming Organs”, “Diseases of the Circulatory System”, “Diseases of the Digestive System”, “Diseases of the Genitourinary System”, “Diseases of the Musculoskeletal System and Connective Tissue”, “Diseases of the Nervous System and Sense Organs”, “Diseases of the Respiratory System”, “Endocrine Nutritional Metabolic Diseases and Immunity Disorders”, “Mental Disorders”, and “Neoplasms”. To ensure the chronological consistency of patient data, individuals whose initial disease diagnosis occurred within their first two clinical visits are excluded from the analysis.

For all models, we randomly partition the dataset into three parts, training data (70%), validation data (10%), and testing data (20%). In our experiments, the best-performing model on the validation data is selected and its performance is further evaluated on the test data in a 3-fold cross validation by calculating the area under a receiver operating characteristic curve (AUC), precision, recall and F1 score.

Baseline Models

We evaluate five baseline models, categorized into two groups. Group 1 consists of models that do not incorporate a disease attention mechanism, including logistic regression (LR), XGBoost²⁴, and recurrent neural networks (RNN). Group 2 includes models that utilize an attention mechanism, such as GRAM²⁵ and Dipole²⁶. Notably, while GRAM leverages disease taxonomy, it relies solely on patients’ historical diagnosis codes as input.

Results

Model Performance

Table 2 shows the performance comparison between ALIGATEHR-Gen and five baseline models. ALIGATEHR-Gen outperformed all baseline methods and achieved an improvement of 6% in the average AUC across 118 diseases compared to the best-performing baseline model (0.76 for ALIGATEHR-Gen vs. 0.72 for GRAM). Additionally, ALIGATEHR-Gen demonstrated the highest average precision, recall and F1 score. Among the three models in group 1, LR and XGBoost did not utilize time-series data as input, whereas RNN did. Despite this, XGBoost outperformed RNN in predictive performance. In group 2, GRAM had slightly better overall performance than Dipole across diseases. By incorporating attention mechanisms into graph-structured data, GRAM assigns varying weights to neighboring nodes, allowing it to prioritize clinically relevant features while minimizing the influence of less significant ones. Moreover, ALIGATEHR-Gen achieved a higher AUC than all five baseline models across all individual diseases (Figure 2). These results highlight the significance of incorporating family health information and medical ontology in improving disease risk prediction. Importantly, ALIGATEHR-Gen dynamically adjusts the importance of clinical features by leveraging both the patient’s health data and that of their first-degree relatives, and demographic information. This leads to improved interpretability and predictive accuracy in medical diagnosis tasks, particularly when handling heterogeneous patient data and capturing long-term dependencies across clinical visits.

Table 2. Comparison of model performance. Average performance on disease risk prediction tasks with 95% confidence interval.

Model	AUC	Precision	Recall	F1
Logistic Regression	0.60±0.05	0.60±0.05	0.58±0.05	0.59±0.05
Recurrent Neural Networks	0.62±0.05	0.56±0.05	0.63±0.05	0.59±0.05
XGBoost	0.68±0.05	0.64±0.06	0.69±0.05	0.66±0.05
Dipole	0.71±0.07	0.72±0.07	0.68±0.07	0.70±0.07
GRAM	0.72±0.06	0.74±0.05	0.68±0.06	0.71±0.06
ALIGATEHR-Gen ^a	0.64±0.05	0.64±0.05	0.61±0.05	0.62±0.05
ALIGATEHR-Gen ^b	0.65±0.06	0.67±0.06	0.61±0.06	0.64±0.06
ALIGATEHR-Gen	0.76±0.06	0.78±0.06	0.71±0.06	0.74±0.06

ALIGATEHR-Gen^a without patient graph.

ALIGATEHR-Gen^b with constant weights on edges of the patient graph.

To evaluate the robustness of our proposed model, we performed an ablation experiment to evaluate the impact of the patient graph on the performance of ALIGATEHR-Gen. Our objective is to determine whether our model effectively captures information from family members who significantly influence a patient’s disease risk. Specifically, we

considered the following setting: 1) ALIGATEHR-Gen^a: the patient graph is removed, and 2) ALIGATEHR-Gen^b: the patient graph is retained, but all edges are assigned constant weights. We observed that removing the patient graph or assigning constant weights to first-degree relatives significantly affected the performance of ALIGATEHR-Gen (Table 2). These results confirm the importance of incorporating first-degree relatives' health information in disease risk prediction. Moreover, merely linking patients to their relatives in a graph is insufficient for optimal performance if attention weights remain constant. This demonstrates that ALIGATEHR-Gen effectively identifies and prioritizes the most influential first-degree relatives for each patient in terms of disease risks.

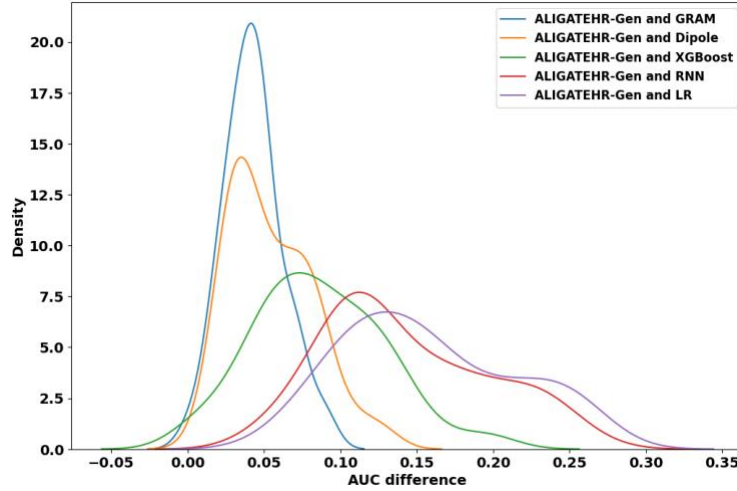


Figure 2. Performance evaluation of ALIGATEHR-Gen using area under the curve (AUC). Kernel density estimate (KDE) of the AUC difference between ALIGATEHR-Gen and five baseline models across diseases.

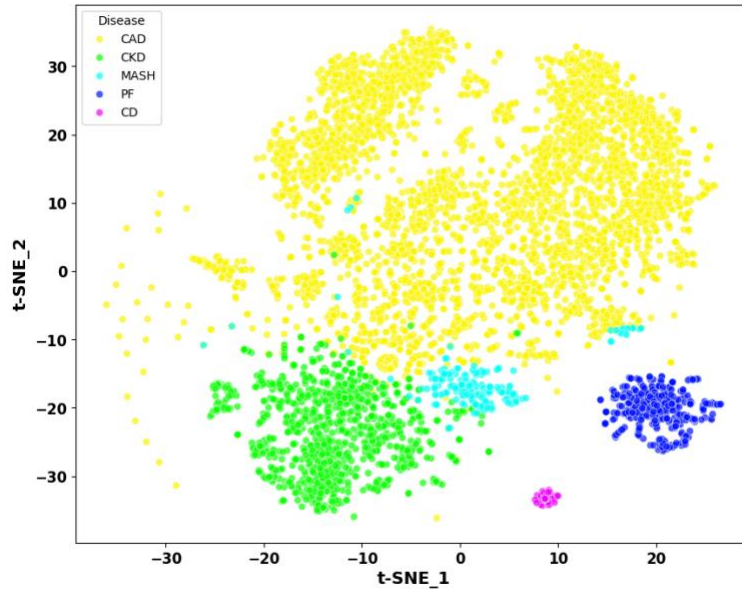


Figure 3. t-SNE plot of patients with one of the five primary fibrotic or closely related diseases using learned latent patient representation, showing the separation and clustering of these diseases.

Case Study

To illustrate the utility of ALIGATEHR-Gen in analyzing individual diseases, we applied it to distinguish among five primary fibrotic or closely related diseases: coronary artery disease (CAD), chronic kidney disease (CKD), metabolic-associated steatohepatitis (MASH), pulmonary fibrosis (PF), and Crohn's disease (CD). Fibrotic diseases, characterized by progressive tissue scarring, frequently occur with comorbidities, complicating both diagnosis and management. The prevalence and public health burden of these conditions are anticipated to increase, largely influenced by strong genetic predispositions and familial risk factors. In this study, we selected five primary fibrotic

or closely related diseases affecting different organ systems to explore common and distinct patient patterns. This analysis aims to provide insights into the genetic influences on fibrotic and related disease risk. We trained a multi-class classifier using ALIGATEHR-Gen to predict the likelihood of each disease in patients diagnosed with exactly one of the five conditions. Table 1 summarizes the demographic characteristics of the patient population for each disease. A t-SNE plot of patient representations learned by ALIGATEHR-Gen reveals a distinct separation between patients with PF and CD from those with the other three diseases (Figure 3). In contrast, CAD, CKD and MASH patients exhibit partial overlap, likely reflecting their shared risk factors and underlying pathophysiological correlations²⁷. These three diseases are largely influenced by metabolic dysregulation, chronic inflammation, and cardiovascular comorbidities, leading to overlapping clinical presentations and pathophysiological mechanisms²⁸⁻³⁰. Additionally, they frequently co-occur in patients with metabolic syndrome, hypertension, and diabetes, further contributing to the similarities in their learned patient representations. This observed clustering pattern suggests that ALIGATEHR-Gen effectively captures the complex interplay among these conditions, reflecting their interconnected etiology and shared genetic or environmental influences. By leveraging these insights, ALIGATEHR-Gen may enhance the understanding of fibrotic disease progression and facilitate more precise risk stratification.

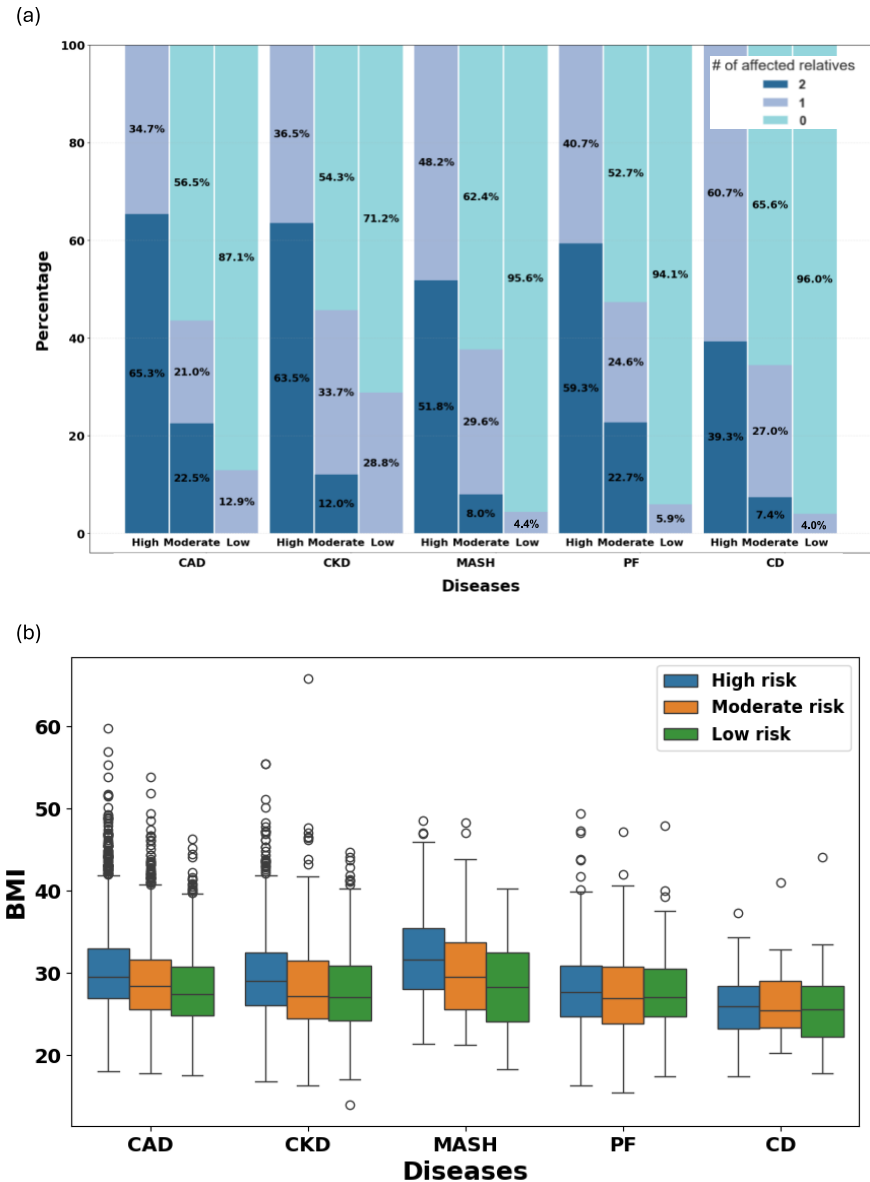


Figure 4. Interpretability analysis of ALIGATEHR-Gen’s predicted risk for the five primary fibrotic or closely related diseases. (a) Association of the number of affected family members and risk groups for each disease. (b) Association between Body Mass Index (BMI) and risk groups for each disease.

Furthermore, we examined the influence of first-degree relatives on the risk of developing each of the five primary fibrotic or closely related diseases. After applying the risk prediction model, patients were stratified into high-risk (predicted risk > 75%), moderate-risk (predicted risk 25-75%), and low-risk (predicted risk < 25%) groups. To assess the familial influences on disease risk, we calculated the number of affected relatives within each risk group. Across all five conditions, high-risk patients consistently had a greater number of affected family members compared to those in the moderate- and low-risk groups, emphasizing the critical role of family health history in fibrotic disease susceptibility (Figure 4a). Since obesity is a common risk factor for CAD, CKD and MASH²⁷, we further examined the relationship between body mass index (BMI) and disease risk, where BMI was not used in the prediction model. As shown in Figure 4b, high-risk patients with CAD, CKD and MASH generally had higher BMI than those in the moderate- and low-risk groups. In contrast, the BMI distributions among patients with PF and CD remained similar across the three risk groups. Notably, patients with MASH had the highest BMI among all five diseases, aligning with obesity as a major risk factor for MASH³¹. This finding highlights the strong association between metabolic dysfunction and fibrotic disease risk³²⁻³⁴, suggesting the need to consider metabolic factors in disease prediction and management.

Discussion

In this study, we present ALIGATEHR-Gen, a deep learning framework that integrates EHR, genetic data, and external medical ontology into an attention-based predictive model. By leveraging genetically inferred first-degree relationships, ALIGATEHR-Gen addresses the underutilization of family health history in disease risk prediction, where such information is often missing or incomplete¹⁰. Our experimental results across 118 diseases in the UKB demonstrate that explicitly modeling genetically inferred first-degree relationships using ALIGATEHR-Gen leads to substantial improvement in both predictive accuracy and model robustness. This improvement is mostly driven by two important factors. First, incorporating genetically inferred relationships captures the latent genetic predispositions that can be overlooked by EHR-centric models, thereby enhancing risk stratification for diseases with strong familial components. Second, the attention mechanisms employed in both the patient graph and medical ontology graph enable ALIGATEHR-Gen to prioritize the most influential first-degree relatives and identify clinically relevant features for improving disease risk prediction.

Our case studies on five primary fibrotic and closely related diseases further highlight the potential of multimodal learning for complex conditions, with or without overlapping risk factors and pathophysiological mechanisms. The identified high-risk patients, characterized by both familial predisposition and elevated BMI, demonstrates ALIGATEHR-Gen's capacity to capture interactions between genetic factors, demographics, and clinical features. These insights can inform targeted interventions, personalized disease management, and improved resource allocation in clinical settings.

The ALIGATEHR-Gen model has several limitations. First, while it focuses on genetically inferred first-degree relationships, it may miss valuable information from more distant relatives, such as second- and third-degree relationships. Accounting for these extended familial relationships could further enhance prediction accuracy and provide deeper insights into the genetic influence on heritable diseases. Second, although kinship coefficients offer a statistically robust approach for inferring family relationships, they do not fully capture shared environmental or lifestyle factors among relatives. Future work could refine the predictive model by including additional measures of shared exposures to better account for these non-genetic influences. Lastly, while disease-specific polygenic risk scores (PRS), which estimate an individual's genetic liability to disease based on their genetic profile and relevant GWAS data, typically explain only a small fraction of phenotypic variation, their integration with family health history and clinical risk factors could further improve disease risk assessment and personalized healthcare strategies.

To better demonstrate the generalizability and scalability of ALIGATEHR-Gen in real-world clinical settings, future work will focus on validating the model in more diverse and heterogeneous populations, such as the All of Us Research Program³⁵ or regional health systems with available genetic data. Such validation will help assess model robustness, fairness across diverse populations, and adaptability to different clinical data systems. Demonstrating consistent performance across populations and clinical environments is important for clinical adoption. Furthermore, successful integration of ALIGATEHR-Gen into clinical decision support systems will require to address several practical challenges, including computational scalability, interoperability with existing EHR systems, and alignment of model outputs with clinical workflows. A key component of this translation involves enhancing model interpretability, providing clinicians with clear and relevant insights into risk prediction and the contributing risk factors, such as specific clinical features or genetic variants. We envision the potential of ALIGATEHR-Gen being deployed as part

of a clinical decision support system that presents risk predictions alongside interpretable factors, empowering clinicians to make informed decisions. Future research will further explore how the model's attention output and relative risk signals can be translated into actionable insights. These efforts are essential to ensure that multimodal disease risk predictive models like ALIGATEHR-Gen not only perform well in retrospective analyses but also provide insights in clinical practice.

Conclusions

In this work, we present ALIGATEHR-Gen, an attention-based graph neural network that integrates EHR, genetic data, and external medical ontology to improve disease risk prediction. Our model achieves superior predictive performance across a broad spectrum of diseases, highlighting the importance of incorporating genetic information in patient representation learning. By leveraging an attention mechanism at both the patient and ontology levels, ALIGATEHR-Gen robustly identifies influential clinical features and first-degree relatives, offering enhanced interpretability and predictive accuracy. Furthermore, this framework effectively addresses practical challenges in clinical settings, such as distinguishing among five primary fibrotic or closely related diseases. Overall, these findings emphasize the value of integrating clinical and genetic data in healthcare practice and highlight the potential of attention-based deep learning in advancing precision medicine.

Ethics Statements

This study uses data from the UK Biobank (project ID: 41910), a large-scale biomedical database and research resource. UK Biobank received research ethics approval from the North West Multi-centre Research Ethics Committee (MREC). All participants provided informed consent at the time of enrollment.

Funding

This work was supported by the National Institutes of Health (NIH) grant R01LM014087.

References

1. Liu J, Cen X, Yi C, et al. Challenges in AI-driven Biomedical Multimodal Data Fusion and Analysis. *Genomics Proteomics Bioinformatics*. Feb 27 2025;doi:10.1093/gpbjnl/qzaf011
2. Acosta JN, Falcone GJ, Rajpurkar P, Topol EJ. Multimodal biomedical AI. *Nat Med*. Sep 2022;28(9):1773-1784. doi:10.1038/s41591-022-01981-2
3. Xu D, Wang C, Khan A, et al. Quantitative disease risk scores from EHR with applications to clinical risk stratification and genetic studies. *Npj Digit Med*. Jul 23 2021;4(1):116. doi:10.1038/s41746-021-00488-3
4. Tsai ML, Chen KF, Chen PC. Harnessing Electronic Health Records and Artificial Intelligence for Enhanced Cardiovascular Risk Prediction: A Comprehensive Review. *J Am Heart Assoc*. Mar 18 2025;14(6):e036946. doi:10.1161/JAHA.124.036946
5. Choi E, Bahadori MT, Schuetz A, Stewart WF, Sun J. Doctor AI: Predicting Clinical Events via Recurrent Neural Networks. *JMLR Workshop Conf Proc*. Aug 2016;56:301-318.
6. Choi E, Bahadori MT, Searles E, Coffey C, Sun J. Multi-layer Representation Learning for Medical Concepts. 2016:
7. Khemani B, Patil S, Kotecha K, Tanwar S. A review of graph neural networks: concepts, architectures, techniques, challenges, datasets, applications, and future directions. *J Big Data-Ger*. Jan 16 2024;11(1)doi:ARTN 18 10.1186/s40537-023-00876-4
8. Yang Z, Mitra A, Liu W, Berlowitz D, Yu H. TransformEHR: transformer-based encoder-decoder generative model to enhance prediction of disease outcomes using electronic health records. *Nat Commun*. Nov 29 2023;14(1):7857. doi:10.1038/s41467-023-43715-z
9. Waqas A, Tripathi A, Ramachandran RP, Stewart PA, Rasool G. Multimodal data integration for oncology in the era of deep neural networks: a review. *Front Artif Intell*. 2024;7:1408843. doi:10.3389/frai.2024.1408843
10. Ginsburg GS, Wu RR, Orlando LA. Family health history: underused for actionable risk assessment. *Lancet*. Aug 17 2019;394(10198):596-603. doi:10.1016/S0140-6736(19)31275-9
11. Huang X, Arora J, Erzurumluoglu AM, et al. Enhancing patient representation learning with inferred family pedigrees improves disease risk prediction. *J Am Med Inform Assoc*. Mar 1 2025;32(3):435-446. doi:10.1093/jamia/ocae297
12. Bycroft C, Freeman C, Petkova D, et al. The UK Biobank resource with deep phenotyping and genomic data. *Nature*. Oct 2018;562(7726):203-209. doi:10.1038/s41586-018-0579-z
13. Shibata K, Hozawa A, Tamiya G, et al. The confounding effect of cryptic relatedness for environmental risks of systolic blood pressure on cohort studies. *Mol Genet Genomic Med*. May 2013;1(1):45-53. doi:10.1002/mgg3.4

14. Voight BF, Pritchard JK. Confounding from cryptic relatedness in case-control association studies. *PLoS Genet.* Sep 2005;1(3):e32. doi:10.1371/journal.pgen.0010032
15. Zhou X, Stephens M. Genome-wide efficient mixed-model analysis for association studies. *Nat Genet.* Jun 17 2012;44(7):821-4. doi:10.1038/ng.2310
16. Loh PR, Tucker G, Bulik-Sullivan BK, et al. Efficient Bayesian mixed-model analysis increases association power in large cohorts. *Nat Genet.* Mar 2015;47(3):284-90. doi:10.1038/ng.3190
17. Vaswani A, Shazeer N, Parmar N, et al. Attention Is All You Need. *Advances in Neural Information Processing Systems 30 (Nips 2017)*. 2017;30
18. Manichaikul A, Mychaleckyj JC, Rich SS, Daly K, Sale M, Chen WM. Robust relationship inference in genome-wide association studies. *Bioinformatics.* Nov 15 2010;26(22):2867-73. doi:10.1093/bioinformatics/btq559
19. Cheesman R, Coleman J, Rayner C, et al. Familial Influences on Neuroticism and Education in the UK Biobank. *Behav Genet.* Mar 2020;50(2):84-93. doi:10.1007/s10519-019-09984-5
20. GenomeAsia KC. The GenomeAsia 100K Project enables genetic discoveries across Asia. *Nature.* Dec 2019;576(7785):106-111. doi:10.1038/s41586-019-1793-z
21. Hofmeister RJ, Ribeiro DM, Rubinacci S, Delaneau O. Accurate rare variant phasing of whole-genome and whole-exome sequencing data in the UK Biobank. *Nat Genet.* Jul 2023;55(7):1243-1249. doi:10.1038/s41588-023-01415-w
22. Hochreiter S, Schmidhuber J. Long Short-Term Memory. *Neural Computation.* 1997;9(8):1735-1780. doi:10.1162/neco.1997.9.8.1735
23. Miotto R, Li L, Kidd BA, Dudley JT. Deep Patient: An Unsupervised Representation to Predict the Future of Patients from the Electronic Health Records. *Sci Rep-Uk.* May 17 2016;6doi:ARTN 26094 10.1038/srep26094
24. Chen TQ, Guestrin C. XGBoost: A Scalable Tree Boosting System. *Kdd'16: Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining.* 2016:785-794. doi:10.1145/2939672.2939785
25. Choi E, Bahadori MT, Song L, Stewart WF, Sun J. GRAM: Graph-based Attention Model for Healthcare Representation Learning. *KDD.* Aug 2017;2017:787-795. doi:10.1145/3097983.3098126
26. Ma F, Chitta R, Zhou JZ, You QY, Sun T, Gao J. Dipole: Diagnosis Prediction in Healthcare via Attention-based Bidirectional Recurrent Neural Networks. 2017:
27. Theodorakis N, Nikolaou M. From Cardiovascular-Kidney-Metabolic Syndrome to Cardiovascular-Renal-Hepatic-Metabolic Syndrome: Proposing an Expanded Framework. *Biomolecules.* Feb 2 2025;15(2)doi:10.3390/biom15020213
28. Villarroel C, Karim G, Sehmbhi M, Debroff J, Weisberg I, Dinani A. Advanced Fibrosis in Metabolic Dysfunction-Associated Steatotic Liver Disease Is Independently Associated With Reduced Renal Function. *Gastro Hep Adv.* 2024;3(1):122-127. doi:10.1016/j.gastha.2023.09.008
29. Targher G, Byrne CD, Tilg H. MASLD: a systemic metabolic disorder with cardiovascular and malignant complications. *Gut.* Mar 7 2024;73(4):691-702. doi:10.1136/gutjnl-2023-330595
30. Ostrominski JW, Arnold SV, Butler J, et al. Prevalence and Overlap of Cardiac, Renal, and Metabolic Conditions in US Adults, 1999-2020. *JAMA Cardiol.* Nov 1 2023;8(11):1050-1060. doi:10.1001/jamacardio.2023.3241
31. Arvanitakis K, Papadakos SP, Lekakis V, et al. Meeting at the Crossroad between Obesity and Hepatic Carcinogenesis: Unique Pathophysiological Pathways Raise Expectations for Innovative Therapeutic Approaches. *Int J Mol Sci.* Sep 28 2023;24(19)doi:10.3390/ijms241914704
32. Nguyen A, Khafagy R, Gao Y, et al. Association Between Obesity and Chronic Kidney Disease: Multivariable Mendelian Randomization Analysis and Observational Data From a Bariatric Surgery Cohort. *Diabetes.* Apr 1 2023;72(4):496-510. doi:10.2337/db22-0696
33. Kreuter M, Bendstrup E, Jouneau S, et al. Weight loss and outcomes in subjects with progressive pulmonary fibrosis: data from the INBUILD trial. *Respir Res.* Mar 9 2023;24(1):71. doi:10.1186/s12931-023-02371-z
34. Gregory AB, Lester KK, Gregory DM, Twells LK, Midodzi WK, Pearce NJ. The Relationship between Body Mass Index and the Severity of Coronary Artery Disease in Patients Referred for Coronary Angiography. *Cardiol Res Pract.* 2017;2017:5481671. doi:10.1155/2017/5481671
35. All of Us Research Program I, Denny JC, Rutter JL, et al. The "All of Us" Research Program. *N Engl J Med.* Aug 15 2019;381(7):668-676. doi:10.1056/NEJMSr1809937